

Limits – 1

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Theoretical minimum.

Def. $x_n \rightarrow L$ means: for every $\varepsilon > 0$ all but finitely many terms lie in $(L - \varepsilon, L + \varepsilon)$. A sequence converges if and only if it is a Cauchy sequence; a bounded monotone sequence converges; every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).

Theorem 1 (Stolz–Cesàro). If (b_n) is strictly increasing and unbounded and $\lim \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$, then $\lim \frac{a_n}{b_n} = L$. In particular the Cesàro means of a convergent sequence converge to the same limit.

Theorem 2 (Fekete’s subadditive lemma). If $x_{n+m} \leq x_n + x_m$ for all $n, m \geq 1$, then $\lim \frac{x_n}{n}$ exists in $[-\infty, +\infty)$ and equals $\inf_{n \geq 1} \frac{x_n}{n}$.

Fact 1. For a recursion $x_{n+1} = f(x_n)$ with continuous f : if the sequence converges, its limit is a fixed point of f ; monotonicity of f makes the sequence monotone on each side of a fixed point, and $|f'| \leq q < 1$ near the fixed point gives geometric convergence.

Fact 2. If $\sum a_n$ converges and (a_n) is monotone, then $na_n \rightarrow 0$: the tail of a convergent series controls the size of its terms.

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1. (Putnam 1969) Let a sequence (x_n) be given and put $y_n = 2x_n + x_{n-1}$ for $n \geq 2$. Suppose that (y_n) converges. Prove that (x_n) converges as well.
 2. Compute $\lim_{n \rightarrow \infty} \left| \sin(\pi \sqrt{n^2 + n + 1}) \right|$.
 3. Let $(x_n)_{n \geq 1}$ be a sequence of reals with $x_{n+m} \leq x_n + x_m$ for all $n, m \geq 1$.
 1. Show that $\lim_{n \rightarrow \infty} \frac{x_n}{n}$ exists and equals $\inf_{n \geq 1} \frac{x_n}{n}$.
 4. Prove that the sequence $a_n = \sqrt{1 + \sqrt{2 + \sqrt{3 + \cdots + \sqrt{n}}}}$, $n \geq 1$, is convergent.

5. Show that if the series $\sum a_n$ converges and (a_n) is decreasing, then $\lim_{n \rightarrow \infty} na_n = 0$.